

 UNIVERSIDAD AUTÓNOMA DE AGUASCALIENTES		Universidad Autónoma de Aguascalientes	Centro de Ciencias Básicas Departamento de Sistemas Electrónicos
		Laboratory:	Sistemas Embebidos
Professor:	Juan Pedro Cisneros Santoyo		SETD
Lab:	5	16×2 LCD in 4-bit mode	

1. Description

- Create a **C library** for a 16×2 HD44780-compatible LCD in **4-bit mode**. The library shall expose an initialization `Init4Bits` and three user functions: `LcdWriteString`, `LcdScrollLeft`, `LcdScrollRight` (and I recommend creating one more `LcdSetCursor`, this function can be implemented inside). Test the library by connecting the LCD to MCU GPIOs and verify writing and scrolling.

2. Objectives

- Implement the initialization sequence for 4-bit mode.
- Provide functions to write strings and scroll the display left and right.
- Produce a modular, reusable API (LCD handle).
- Verify on hardware and document usage.

3. Materials & Tools

- STM32 (or similar) and Keil.
- LCD 16×2, pot for contrast, resistors, wires.
- Recommendation: tie RW to GND. Use RS, EN, D4..D7 signals.

4. Development steps

1. Hardware wiring (Vcc, GND, V0 pot, RS, EN, D4..D7, RW=GND).
2. Define `LCD_HandleTypeDef` describing pins/ports.
3. Implement `LcdInit4Bits`, `LcdWriteString`, `LcdScrollLeft`, `LcdScrollRight`.
4. Test with short and long strings and use scroll functions.
- 5.

5. Important notes / possible improvements

- To avoid slow fixed delays, connect RW to MCU and implement busy flag polling. That improves performance but increases complexity.
- If you use an RTOS, protect the LCD functions with a mutex.
- `LcdScrollLeft/Right` use the display SHIFT command; for more complex text scrolling you might implement a buffer + redraw.

6. Suggested tests

- Initialize and write "Hello World".
- Write Line1\nLine2.
- Test scrolling a long string with lcdScrollLeft(..., times, delay_ms).

7. Expected results

- Correct 4-bit initialization.
- Correct text writing and line switching.
- Smooth visible scroll via display shift commands.

8. Reflection / evaluation questions

- Explain the initialization 0x3,0x3,0x3,0x2 sequence.
- Why tie RW to GND or why not? Tradeoffs.
- How would you generalize to 20×4 displays?
- How to avoid message interleaving when multiple tasks print?

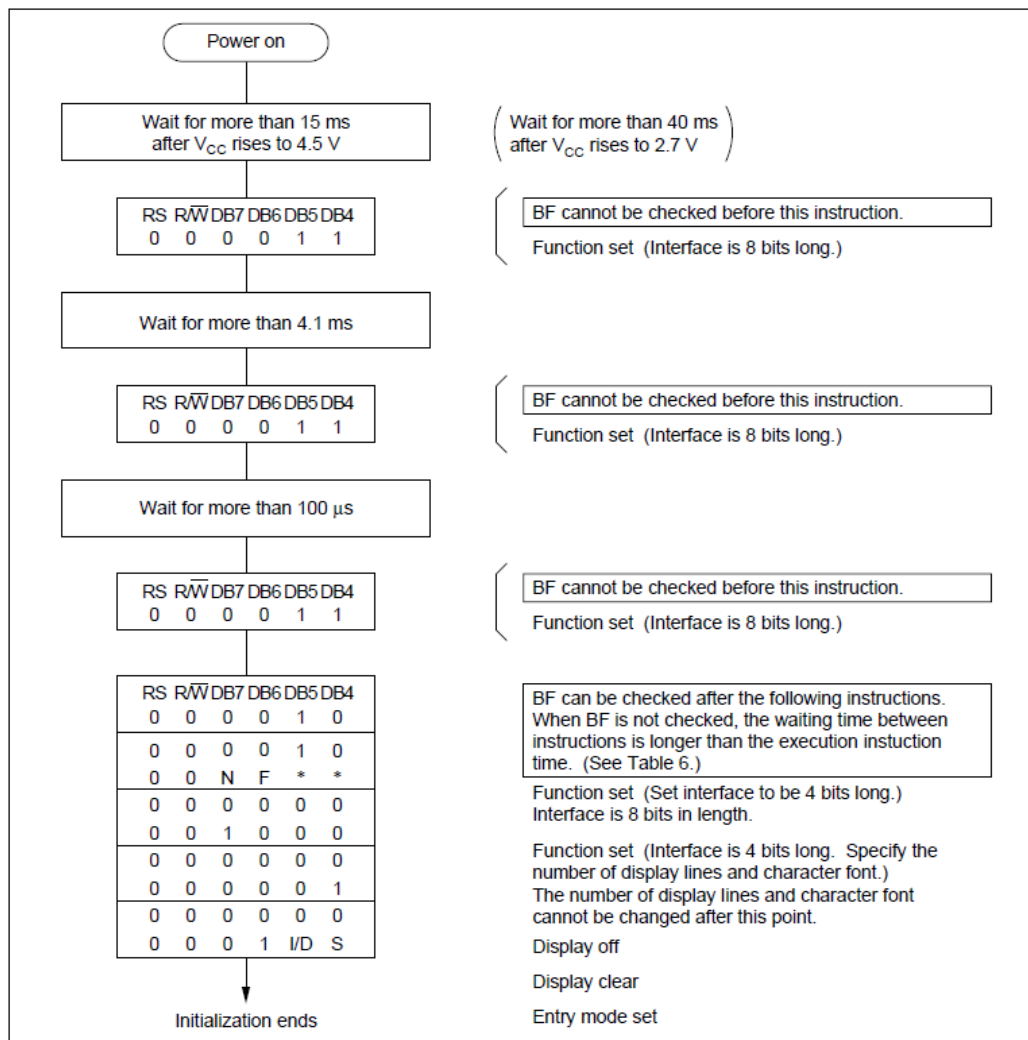


Figure 24 4-Bit Interface